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Air Supplement as a Stimulation Approach for the In Situ Desulfurization and Methanization Enhancement of Anaerobic Digestion of Chicken Manure

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ABSTRACT: Anaerobic digestion of chicken manure produces biogas with an extreme H_2S content (5000−6000 ppm), which greatly weakens the economic viability of biogas plants. In this study, in situ desulfurization through supplying limited air into the headspace of a digester and the impact upon the process performance were evaluated under different oxygen loads (1.4, 2.8, and 4.2 mL/gVS_{in}) and injection approaches (single pulse and continuously). The results showed that an oxygen load at 4.2 mL/gVS_{in} with continuous injection removed 99.7% of H_2S (reduced to 10−15 ppm) and simultaneously increased the methane yield by 6.4%. Sulfide-oxidizing bacteria enrichment and the increased concentration of soluble iron were the main features of the air supplement system. The 16S rDNA results explained the attached growth of sulfur-oxidizing bacteria on the headspace carriers and the enrichment of *Methanobacterium* and *Methanoculleus* under the microaerated system. The obtained results suggest that in situ desulfurization technology could be applied more widely.

1. INTRODUCTION

Anaerobic digestion (AD) is a promising solution for stabilizing organic waste, reclaiming bioenergy, and producing digestate as organic fertilizer. In 2019, China issued a policy aiming to accelerate the development of bionatural gas, through the AD of waste, to 20 billion m^3 by the end of 2030 (www.nea.gov.cn). In China, around 155 million tons of chicken manure are produced annually, and regarding its higher potential biogas production (794 mL/gVS),[1] chicken manure is deemed to be a suitable substrate for AD treatment. In addition, recent studies indicated that the AD of chicken manure was successful even under stressed conditions such as high ammonia level.[2,3] However, AD of chicken manure generates biogas with H_2S as high as 5000−6000 ppm, which is much higher than that produced from the AD of diary manure (2000−4000 ppm) and sewage sludge (2500−4900 ppm).[4,5] Biogas containing high concentrations of H_2S could corrode gas pipes, mechanical devices, and engines, and thus, it must be purified before its further utilization.[6] A wide range of desulfurization techniques have been applied, but the significant drawbacks with maintenance complexity, high cost, and the secondary pollution have restricted their application.[7] So far, a cost-effective approach for removing H_2S from the AD of chicken manure has remained a challenge.

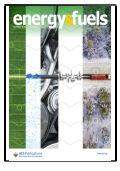
In situ desulfurization technology is proposed as a way of removing H_2S by supplying a limited amount of air/oxygen into the headspace of an anaerobic digester in which the growth of sulfide oxidizing bacteria (SOB) is sustained. Under microaerated conditions, SOB biologically oxidizes H_2S to sulfur or sulfate, depending on the O_2 dosage (in eqs 1 and 2).[8] A wide range of in situ H_2S removal efficiencies have been reported for dairy manure (68%; from 4000 to 1120 ppm),[4] agricultural waste (88%; from 2500 to 300 ppm),[9] and synthetic waste (99%; from 680 to 10 ppm).[10] Recently, H_2S removal from biogas of chicken manure through in situ desulfurization technology was reported in a batch experiment.[11] However, the performance of the in situ H_2S removal in a long-term AD of chicken manure has been scarcely studied, and the underlying mechanism is still unclear. On the other hand, improper air dosage may not only result in a lower H_2S removal efficiency but also increase the risk of disturbing the anaerobic environment.[12] Balancing the air dosage for simultaneous H_2S removal and an efficient AD performance is therefore still a challenge. In addition, a stable habitat with available wetted area is necessary for the biofilm activity, guaranteeing the efficient H_2S removal.[13] However, hitherto, there is little research about the installation of carrier for biofilm growth during the in situ desulfurization process, and the microbial features of the biofilm are still lacking and not quite clear.

$$2HS^- + O_2 \rightarrow 2S^0 + OH^- \quad \Delta G = -169.35 \text{ kJ mol}^{-1} \tag{1}$$

$$2HS^- + 4O_2 \rightarrow 2SO_4^{2-} + 2H^+ \quad \Delta G = -732.58 \text{ kJ mol}^{-1} \tag{2}$$

In addition to this, introducing a small amount of air into the anaerobic digester could eliminate the sulfide in the liquid

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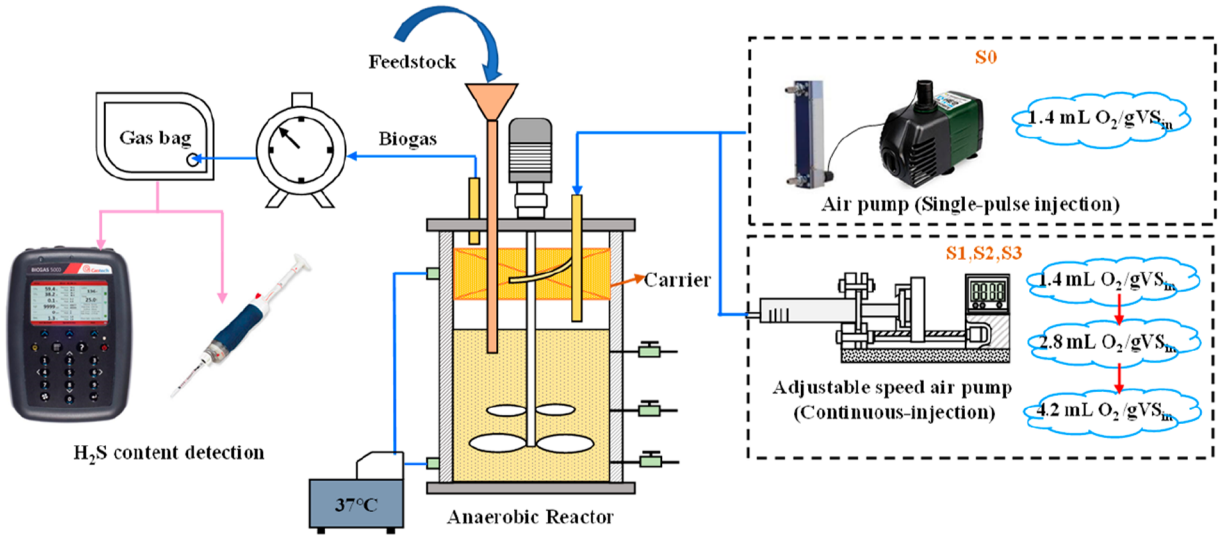


Figure 1. In situ biological desulfurization diagram.

phase, and thus the toxic effects of sulfide on methanogens could be alleviated.[14] Alongside this, the removal of sulfide in the aqueous phase is assumed to hinder the formation of FeS and FeS_2 precipitation. This leads to the release of soluble iron, thus improving its potential bioavailability, which has essential effects on the corresponding enzymes and metabolism of methanogens.[15] So far, though, research into the positive effects of the air supply on the release of soluble iron in the anaerobic digesters is lacking.

In addition, it has been reported that air supplement could significantly enhance the diversity and activity of facultative bacterial communities in the AD system such as some hydrolytic bacteria, that is, Firmicutes, which could in turn promote the overall efficiency and robustness of the AD process.[16,17] The dominant methanogens group has also been found to shift when air was supplied.[5] It has been reported that microaeration caused an increase in the *Methanoculleus* population,[18] while Fu et al. in 2016[19] found that the genus *Methanoculleus* decreased by 43.5% under microaeration conditions. Theoretically, H_2S is generated from the hydrolytic degradation of proteinaceous materials or the sulfate reduction process performed by sulfate reducing bacteria (SRB).[20] It was supposed that the in situ removal of H_2S may be performed through two possible pathways: suppressing the sulfate reduction and/or achieving the H_2S oxidation in the digester headspace.[21,22] So far, the discussions about both mechanisms are limited and controversial; therefore, more knowledge about the specific bacteria group involved in above two pathways is needed. Furthermore, it is worth investigating the hydrolysis, methane production, and H_2S removal process and their associated microorganisms and mechanisms simultaneously for further process optimization.

Based on the knowledge obtained from previous batch experiment, this study, therefore, aimed to evaluate an in situ biological desulfurization strategy using continuously stirred tank reactors (CSTR) treating chicken manure under different air dosing conditions. Attention was similarly focused on more accurately optimizing H_2S exclusion and methane production enhancement simultaneously. The correlation between AD process parameter indicators was investigated by principal component analysis (PCA). The microbial community changes were also investigated through 16S rDNA to further understand the underlying mechanisms of the H_2S removal and improved AD performance.

2. MATERIALS AND METHODS

2.1. Substrates and Inoculum. Raw chicken manure with a total solid (TS) content of 20%−25% was obtained from a chicken farm located in Beijing. The raw chicken manure was stored in a freezing room at 4 °C until use. Feathers and sands were removed manually. The raw chicken manure was diluted, with water, to a slurry with a TS of around 10%. It was then put into a shredder (Joyoung-JCLC012, China) before feeding. The volatile solid (VS) and total chemical oxygen demand (TCOD) of the feeding material were 70 and 101 g/L, respectively. The inoculum was collected from a lab-scale chicken manure digester which was operated for more than 2 years under 37 °C.

2.2. Digesters Configuration and Operation Strategy. The experiment was conducted with two lab-scale CSTRs under mesophilic conditions, with 10 and 15 L working and total volumes, respectively. The digesters were equipped with autocontrolled motors which work at a speed of 150 rpm. The hydraulic retention time (HRT) was set at 40 days, i.e., daily feeding of 250 mL feedstock. The organic loading rate (OLR) was fixed at 1.75 gVS $(L{\cdot}d)^{-1}$. The system diagram is depicted in Figure 1.

The two digesters were started and operated in parallel: R1 as control and R2 as an air supplied digester. Air supplement was applied to R2 at four stages with different injection approaches, namely S0, S1, S2, and S3. The duration of each stage was 30 days, and the total experiment lasted for 120 days. R2 was equipped with an air injection pump with adjustable flow rate (0−500 mL/d). A carrier for biofilm growth was installed in the R2 headspace near the liquid (5 cm from the gas−liquid interphase). The carriers were made of nylon rope and settled with sheet structure by twining around the digester headspace, occupying one-third of the headspace volume. The diameter of the carrier was the same as for the digester (20 cm). Before the experiment started, the carrier was installed in the digester headspace with spraying sludge on that and injecting a mixture of air and H_2S-rich biogas for SOB group acclimatization and biofilm culturing. Meanwhile, some additional parts of carrier were dipped in liquid phase to guarantee the transfer of nutrients and moisture to the carrier.

The initial air dosage was determined by the H_2S content in the biogas generated from R1 according to the theoretical O_2/H_2S ratio (0.5) for H_2S oxidation, as shown in eq 1. During the S0 stage (day 1−30), the air was manually injected into R2 at an oxygen load of 1.4 mL/gVS by using the single-pulse (120 mL/d). At the subsequent S1, S2, and S3 stages, the air was continuously injected at oxygen loads of

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Table 1. Setup of Digesters at Different Stages

			R2			
parameter	units	R1	S0	S1	S2	S3
air injection			single-pulse	continuously		
air dosage	mL/d		120	120	240	360
air flow rate	mL/h			5	10	15
O_2 load	mL/gVS		1.4	1.4	2.8	4.2
HS volume[a]	L	5	5	5	5	5
O_2/H_2S	mol/mol		0.5	0.5	1.0	1.5
duration	d	1−120	1−30	31−60	61−90	91−120

[a]HS: headspace of digester.

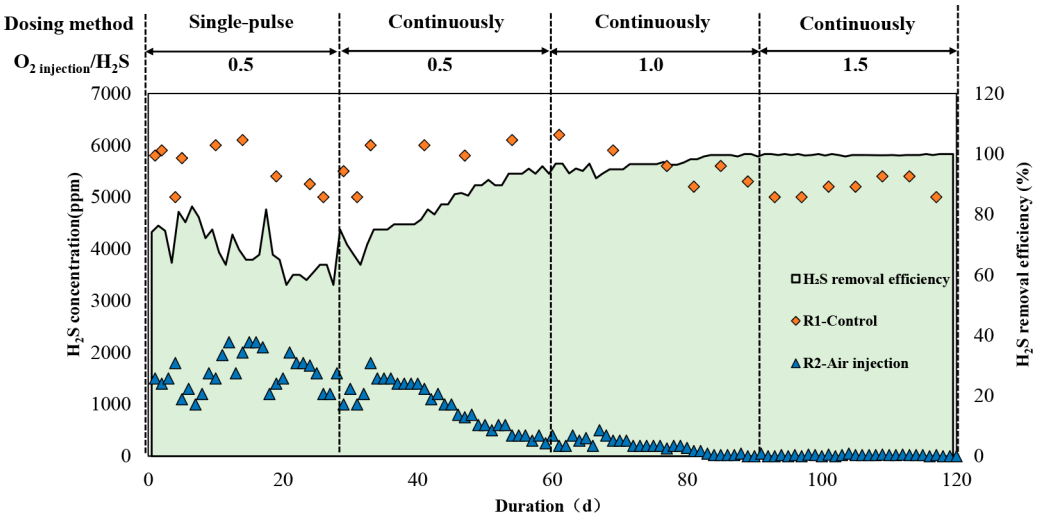


Figure 2. H_2S content and H_2S removal efficiencies in both digesters.

1.4, 2.8, and 4.2 mL/gVS, respectively, through the air pump. Correspondingly, the air flow rates were 5.0, 10.0, and 15.0 mL/h with air amounts of 120, 240, and 360 mL/d. The details about the operation parameters of R1 and R2 are shown in Table 1.

2.3. Chemical Analysis Methods. The biogas generated was collected by a gasbag. The biogas volume was measured by a gas flow meter (LML-1, Jinzhiye Co., LTD, China) and recorded after calibration under standard conditions. The gas composition was measured with a gas chromatograph (SP-2100, Beifen-Ruili Analytical Instrument, China) according to a previous study.[23] The H_2S concentrations were measured daily by using a gas detection tube (0−4000 ppm, Gastec, Japan) and gas composition analyzer (0−1000 ppm, ES-5K, Geotech, UK) from the bag, before the measurement of the daily gas volume at a high and low levels of H_2S, respectively. The pH was determined by a Mettler-Tolendo pH meter, and the above parameters were measured daily. The digester effluent was filtered through a 0.22 μm filter after centrifugation at 8000 rpm for 20 min (Cence TGL-16M, China) for volatile fatty acids (VFAs) and soluble iron analysis. The TS, VS, total chemical oxygen demand (TCOD), and ammonium (NH_4^+-N) were measured according to the American Public Health Association.[24] The compositions of VFAs including acetate, propionate, isobutyrate, butyrate, isovalerate, valeric acid, and caproate were measured every 3 days by a gas chromatograph (Shimadzu GC-2012 plus, Japan). The concentration of soluble iron in effluent was monitored by an inductivity coupled plasma mass spectrometer (ICP-MS-X2, Thermo Fisher, Waltham, MA) as described previously.[25] Statistics significance was obtained by using one-way analysis of variance (ANOVA, α = 0.05).The H_2S concentration in R1 (control) was regarded as a baseline to calculate the removal efficiency in R2 according to eq 3.

$$H_2S \text{ removal efficiency } (\%) = \frac{C_0 - C_d}{C_0} \quad (3)$$

C_d is the H_2S content in R2 (biogas after desulfurized), ppm; C_0 is the H_2S content in R1(control), ppm.

The first-order kinetic model was assigned to simulate the daily methane production as eq 4.

$$\ln(C_{s_0}/C_s) = kt \quad (4)$$

C_{s_0} refers to the highest methane production (mL/gVS), C_s stands for the cumulative methane production of the substrate (mL/gVS), k is the rate constant of the first order ($d - 1$), and t is the lasting time (h).

2.4. Statistical Analysis. In this study, the majority of the analyses were carried in triplicate, and the results were expressed as mean value ± SD (standard deviation). The correlation relationship between the AD process parameter indicators was performed by the PCA method employing the SPSS statistic 25 software package, and the associated PCA bioplot was drew by Origin Pro 2016 software.

2.5. Microbial Community Analysis. Molecular analysis samples were collected on day 120, including the samples from the liquid phase in both digesters and the carrier for the microorganisms in R2; all samples were collected in duplicate and subsequently stored at −20 °C until tested.[2] The method of cetyltrimethylammonium bromide/sodium dodecyl sulfate was assigned for extraction. Forward primers (338F and 524F) and reverse primers (806R and 958R) for bacterial and archaeal sequences, respectively, were performed for amplifications of 16S rDNA. The key characteristics of PCR thermal cycling were taken from a previous study.[3] PCR products were then purified, quantitated, and paired-end-sequenced based on the platform of Illumina MiSeq Ultimately, to remove the poor-equality sequences, the Majorbio I-Sanger Cloud software (www.i-sanger.com) was assigned to analyze the primary data. The sequences were rarefied to 25000 sequences per sample, and those with 97% similarity were integrated into operational taxonomic units (OTUs). Taxonomic

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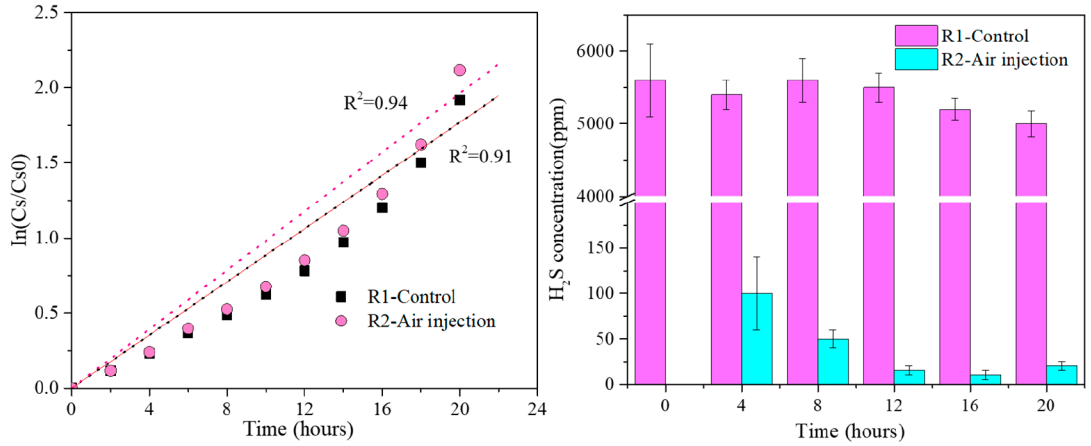


Figure 3. First-order kinetics of methane production (a) and H_2S concentration variation (b) for 20 h at the S3 stage.

Table 2. Performance of the Air Injection under Different Approaches

parameters	units	S0	S1	S2	S3
TS_{in}	g L^{-1}	102 ± 3	104 ± 5	101 ± 4	103 ± 6
VS_{in}	g L^{-1}	72 ± 2	73 ± 3	71 ± 2	73 ± 1
OLR	gVS L^{-1} d^{-1}	1.8 ± 0.4	2.1 ± 0.6	2.0 ± 0.3	2.1 ± 0.1
pH		7.97 ± 0.21	7.98 ± 0.07	7.96 ± 0.04	7.93 ± 0.22
CH_4 yield	mL gVS d^{-1}	329 ± 28	360 ± 13	363 ± 18	367 ± 14
CH_4	%	61.6 ± 0.03	66.1 ± 0.03	64.3 ± 0.02	64.5 ± 0.02
CO_2	%	38.4 ± 0.02	33.9 ± 0.01	35.7 ± 0.02	35.5 ± 0.03
total VFAs	mg L^{-1}	787 ± 179	494 ± 138	488 ± 46	431 ± 62
H_2S removal	%	68 ± 7	86 ± 8	97 ± 2	99 ± 1
CH_4 yield enhancement	%		4.4 ± 0.1 ($P < 0.05$)	5.2 ± 0.2 ($P < 0.05$)	6.4 ± 0.1 ($P < 0.05$)
residual O_2	%	<0.1	<0.1	0.2	0.4

classification was determined by the assigning of a 16S rDNA reference database.

3. RESULTS AND DISCUSSION

3.1. H_2S Removal Efficiency with Different Air Injection Strategies. In the R1 (control), the H_2S concentration in biogas was 5500 ± 400 ppm, which was comparable to previously reported values in chicken manure digesters.[26] This value was utilized to determine the air injection dosage in R2. Initially (at S0 stage), air was supplied by injecting air at 120 mL/d by a single pulse. The H_2S concentration in biogas decreased dramatically to around 1500 ppm, representing a H_2S removal efficiency of 68%. The lower H_2S removal efficiency obtained at the S0 stage may be attributed to the partial consumption of oxygen by other facultative microorganisms instead of SOB, resulting in the insufficient oxygen for subsequent H_2S removal.[27] It was anticipated that the removal efficiency could be improved when matching the air injection with the H_2S generation rates. Hence, from the 31th day (at S1 stage), the same daily amount of air was continuously injected with a flow rate of 0.08 mL/min (5 mL/h). The H_2S removal efficiency attained consequently increased to 86% (Figure 2).

To improve the O_2 availability for SOB and then further enhance the H_2S removal, the air flow rate was increased to 10.0 and 15.0 mL/h during the S2 and S3 stages. Consequently, the H_2S removal efficiency reached 97% and 99.7% in S2 and S3, respectively. The H_2S content in biogas and methane production was measured every 4 and 2 h after feeding, respectively, lasting for 20 h. As shown in Figure 3b, it is obvious that the H_2S content in biogas maintained a low level (<100 ppm) after feeding and sustained to decrease, finally, to <20 ppm. It verified the hypothesis that higher availability of O_2 from increased air dosage for the SOB group significantly improved the efficiency and stability of the H_2S removal process. In practice, to maintain the feasibility of subsequent utilization and reduce the risk of explosion because of the mixture of methane and oxygen, it is important to note that resident O_2 in biogas is undesirable. Therefore, the resident O_2 in biogas at different stages was measured. In Table 2, the O_2 content in biogas was almost negligible and was far from the flammable range of 5−15% of methane.[28]

3.2. Enhancement of Methane Production with Air Injection. The cumulative methane production in R1 and R2 is shown in Figure 4a. The average cumulative methane production of R1 at each stage is 186 L, while that of R2 is 176, 194, 196, and 198 L at the S0, S1, S2, and S3 stage, respectively. In addition, calculated by characteristics of feedstock, the mean methane yield of R1 was 345 ± 12 mL/gVS_{in}, and the methane yield of R2 at the S0 stage was 329 ± 28 mL/gVS_{in} (Table 2). The decreased methane yield was also reported when the chemically precipitated primary sludge was treated with a semicontinuous injection at 500 mL of air/d.[29] The relatively lower methane yield may be ascribed to the unbalanced air injection. The methane yields in R2 with continuous air injections increased to 360 ± 13, 363 ± 18, and 367 ± 14 mL/gVS_{in} during the S1, S2, and S3 stages. They were 4.3% ($p < 0.05$), 5.3% ($p < 0.05$), and 6.4% ($p < 0.05$)

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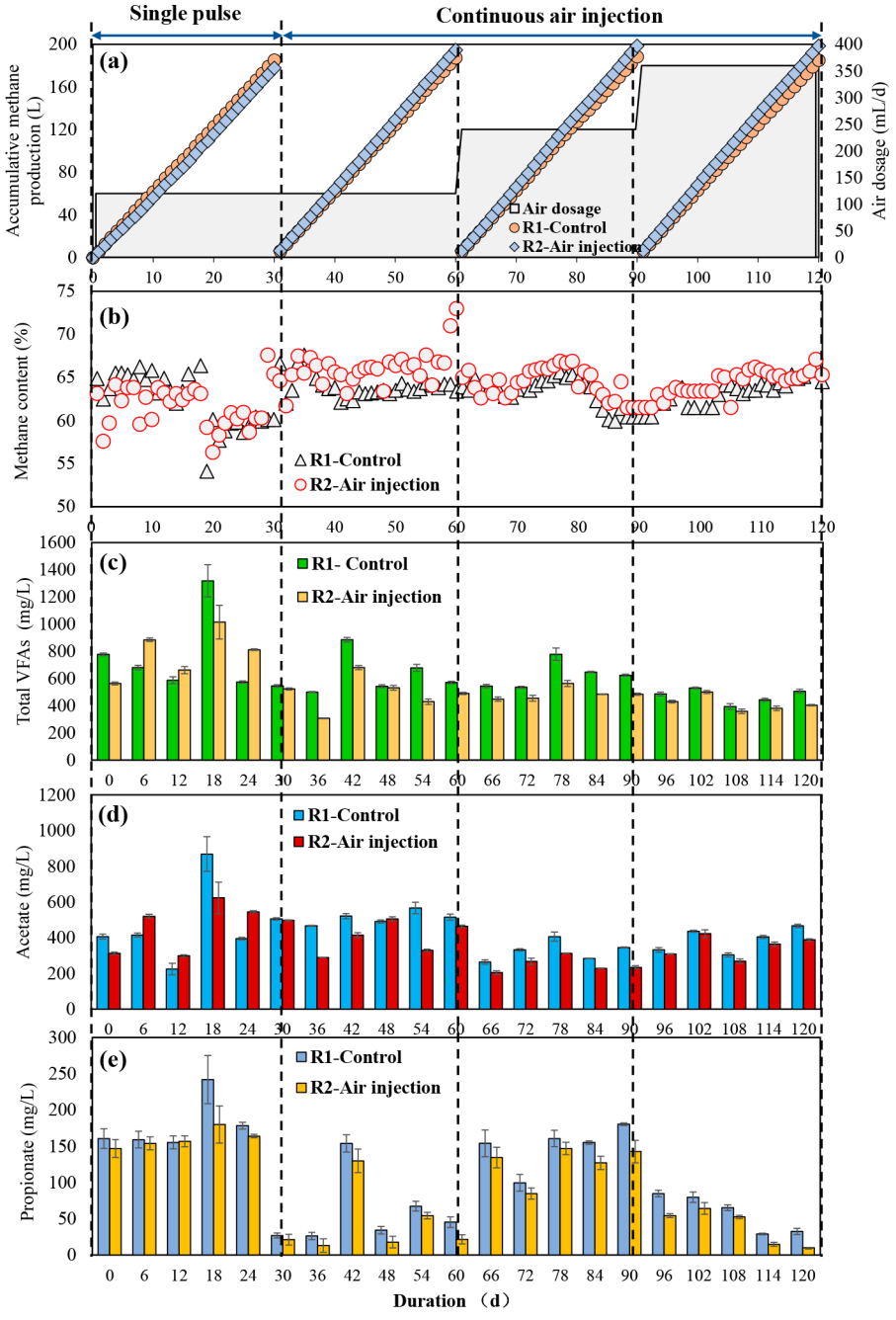


Figure 4. Performance in the R1 control and R2 air injection digester during different stages: (a) accumulative methane production, (b) methane content, (c) total VFAs, (d) acetate, and (e) propionate.

Table 3. Effect of Air Injections on AD Performance and H_2S Removal

materials	reactor type	experiment (days)	air dosage	H_2S removal efficiency (%)	enhancement in methane yield (%)	ref
corn straw	batch	60	5−40 mL/gVS_{in}	not investigated	5.21−16.24	34
food waste	batch	50	37.5 mL/L/d	not investigated	23.25	31
wheat straw	batch	35	5 mL/gVS_{in}	not investigated	7.2	32
waste activated sludge	semicontinuous reactor	210	10 mL/d	>99	not investigated	22
pulp mill wastewater	UASB (10.5 L)	140	3−6 mL/L/min	>30	not investigated	35
primary sludge	CSTR (50 L)	60	0.7 mL/s	99	not investigated	21
sewage sludge and food waste	CSTR (4500 m^3)	560	0.09−0.77 kNm^3/d	>99	not investigated	36
chicken manure	batch	35	7−30 mL/gVS	58	39−77	11
chicken manure	CSTR (10 L)	120	7−21 mL/gVS_{in}	>99	4.4−6.4	this study

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higher than those in the control. A previous investigation reported a slight decrease or negligible increment of methane yield with injecting air with flow rate of 20 m^3/d (~5% of biogas production).[4] To further elucidate the improved AD performance obtained in this study, the first-order kinetics model was assigned to simulate the biogas production in R1 and R2. The methane production rate in R2 at S3 had a higher K_1 (0.098) than that (K_2 = 0.088) in the control (Figure 3a). The raised methane production rate in R2 may be ascribed to the acceleration of facultative microorganisms with the existence of oxygen and the relieving of H_2S on microorganisms.[30] In addition, Figure 4c illustrates that the VFAs in R2 maintain a lower level than that of R1. More specifically, the mean VFA concentrations were 494, 485, and 430 mg/L in R2 during the S1, S2, and S3 stages, which demonstrated a remarkable 21.1%, 22.6%, and 31.5% reduction compared to that of R1 (626 mg/L). Also, the acetate and propionate content in R2 maintained a lower level compared to that in R1 (Figure 4d,e). The acetate in R2 constituted a larger proportion of the total VFAs at a higher oxygen load (4.2 mL/gVS_{in}) compared to R1 (78.9% vs 68.1%); this was also consistent with the previous study, which may be ascribed to the acceleration of degradation and oxidation of unexpected VFA to acetate under microaeration conditions.[31,32] The lower VFA level and absence of longer fatty acids accumulation, that is, propionate, were reported to improve the AD process stability and higher methane yield.[33] Table 3 summarizes studies on H_2S removal efficiency and methane yield enhancement.

Considering the ionization equilibrium of H_2S, HS^-, and S^{2-}, the removal of H_2S from the biogas would therefore reduce the free S^{2-} concentration in the liquid phase. Consequently, the Fe^{2+} which was previously precipitated by S^{2-} may be "released". The soluble iron concentration in the two digesters is shown in Figure 5. With the improvement of

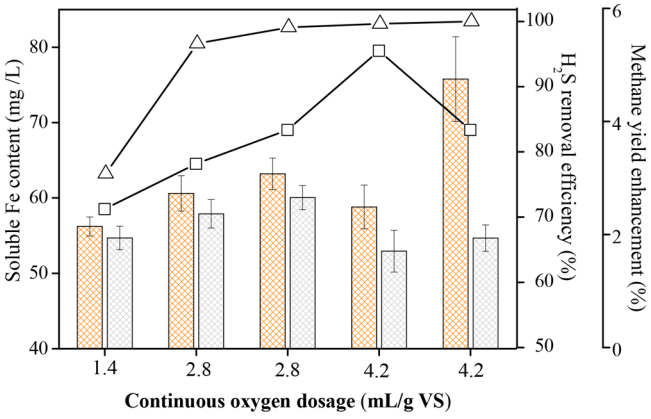


Figure 5. Soluble iron content variation in the R1 (gray bars) and R2 (red bars), H_2S removal efficiency (△), and methane yield enhancement in R2 (□) under different air injection conditions during the research.

H_2S removal efficiency, the soluble iron in R2 became significantly higher than that in R1. During the S3 stage, the soluble iron in R2 increased to 75.8 mg/L, which was higher by 1.4-fold than that of R1 (54.7 mg/L).The raised soluble iron content in R2 may be ascribed to the mitigation of sulfide in the liquid phase, which blocks precipitation of FeS and "releases" free iron, improving the bioavailability of iron for the microorganism. It was reported that addition of a minute amount of Fe^{2+} or Fe^{3+} has a significantly positive effect on the AD performance of swine and dairy manure.[37,38] The raised soluble iron content attained in this study has a positive effect on the methane yield, as shown in Figure 5.

3.3. Correlation between System Operation Parameters and Air Injection. In Figure 6, the PCA was employed

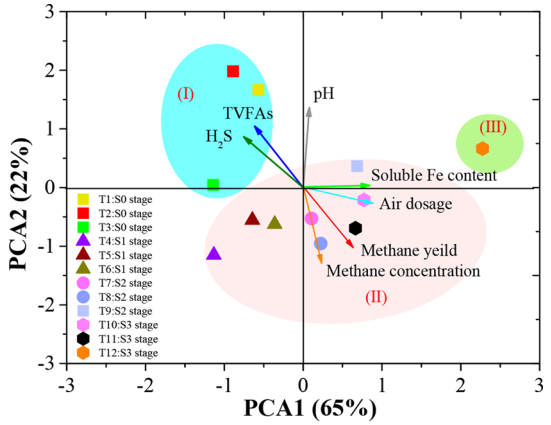


Figure 6. PCA on relationship between process parameter indicators at different stages.

to assess the potential correlation among the parameters. In total, air injection dosages and other AD process parameters were collected at different stages (S0−S3) with various air strategies. Obviously, the two axes in the PCA plot were noted as PCA1 and PCA2, annotating 65% and 22% of the total variance of related data, respectively. PCA plot indicates that the 12 samples were clustered in three areas: (1) T1, T2, T3 (S0), (2) T4−T11 (S1−S3), and (3) T12 (S3). This suggests that the air injection strategy before entering the optimized stage significantly affected the H_2S content in biogas and VFAs level since the two parameters were clustered in the top-left quadrant in the PCA bioplot. However, after shifting to continuous air injection, higher air injection dosages (T7−T12 at S2 and S3) have significantly positive effects on methane yield, soluble iron concentration, and methane content, whereas pH, VFAs, and H_2S were discrepant from the direction of raised air dosages. The presentation of PCA plot was in tangent with the same tendency in Figures 2 and 4. The raised air dosage could improve the AD performance and stability, which may be ascribed to the remission of inhibition of sulfide on the microbial community and enhancement of hydrolytic bacteria under microaeration conditions, degrading more complex substrates to VFA and subsequently converting more VFAs to methane. The results obtained evidenced that the most significant factor is air injection dosage, which directly determines the H_2S removal efficiency and lower level of VFAs, consequently reflecting on enhancement of methane yield. Furthermore, the PCA results illustrate that continuous air injection with slightly excessive air density was more beneficial for AD performance.

3.4. Sulfate Reducing Bacterial Population Structure. The SRB population in the R1 and R2 during the S3 stage was evaluated. The 16S rRNA gene sequencing results indicated that several genera of SRB were identified in both R1 and R2 at relatively low levels. The relative abundance and metabolism characteristics of this SRB group are listed in Table 4.

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Table 4. Diversity of SRB and Characteristics of the Identified SRB Group at the 120th Day

SRB genera	relative abundance (%)			SRB characteristics		
	R1	R2	R2 carrier	type[a]	electron donors	organic oxidation
Desulfurovibrio	0.58	0.28	0.16	G−	H_2 and lactate	incomplete
Desulfobulbus	0.22	0.08	0.05	G−	H_2 or ethanol	incomplete
Desulfitobacterium	<0.01	<0.01	<0.01	G−	H_2 and lactate	incomplete
Desulfuribacillus	0.10	0.14	<0.01	G−		
Sulfurospirillum	<0.01	<0.01	<0.01			

[a]G−: Gram-negative bacteria.

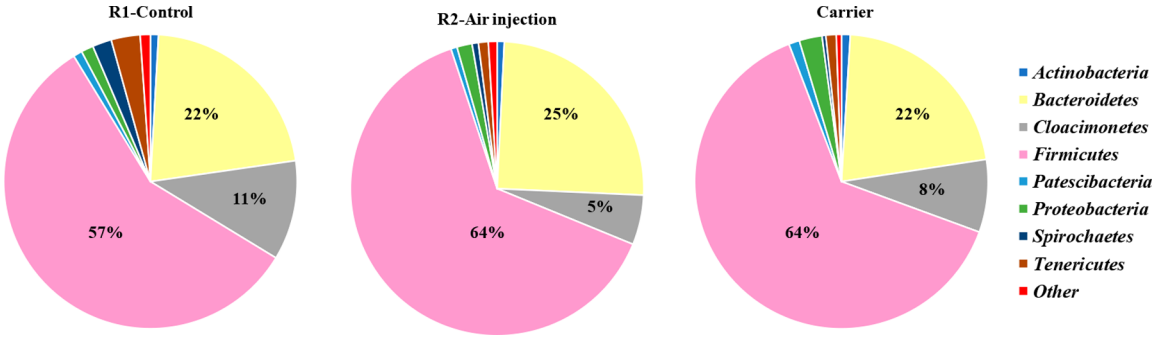


Figure 7. Microbial community structure at the phylum level of bacteria on the 16S rDNA in the R1, R2 digester, and R2 carrier at the 120th day.

Table 5. Diversity and Characteristics of the Identified SOB Group at the 120th Day

SOB genera	relative abundance (%)			electron donors	type
	R1	R2	R2 carrier		
Acinetobactor	0.24	0.52	0.76	S^{2-}, S^0, $S_2O_3^{2-}$	chemolithotrophic
Sulfuricurvum	0.51	0.63	0.96	S^{2-}, S^0, $S_2O_3^{2-}$	chemolithotrophic
Thiobacillus	0.25	0.62	0.78	S^{2-}, S^0, $S_2O_3^{2-}$, $S_4O_6^{2-}$	chemolithotrophic, autotrophic denitrifying
Pseudomonas	0.11	0.12	0.23	S^{2-}, S^0, $S_2O_3^{2-}$, $S_4O_6^{2-}$	chemolithotrophic

All the SRB members identified in both digesters were Gram-negative and belonged to the *Deltaproteobacteria* class. It was observed that genera *Desulfurovibrio* and *Desulfobulbus* were the dominant group among the identified SRB community in both digesters. These two genera exist commonly in anaerobic wastewater treatment systems, and some of their species can reduce SO_4^{2-} and SO_3^{2-} by using hydrogen, lactate, and ethanol as an electron donor.[39] With the air supplement in R2, these two genera were observed to exist rarely with relative abundance of 0.28% and 0.08%, respectively. The genus *Sulfurospirillum*, which was capable of reducing sulfur,[40] was also detected although with scarce existence (<0.01%).

3.5. Bacterial Community in the SOB Carrier and Liquid Phase. The SOB team is commonly known to be the main contributors to biological H_2S oxidation. For instance, Jenicek et al. in 2011[41] demonstrated that SOB group activity was greatly enhanced with the existence of O_2, and it helped attain 99% removal of H_2S. To identify the SOB group responsible for H_2S oxidation in this study, the biosamples were gained from the liquid phase and biofilm of the carrier at the 120th day (S3). As shown in Figure 7, on the carrier, the phylum Firmicutes dominated among the members with a level of 64% while phylum Proteobacteria and Actinobacteria were observed to occupy 2.8% and 1.2%, respectively. It was observed that the majority of SOB members belonged to phylum Proteobacteria and Actinobacteria with genera *Halothiobacillus* and *Sulfuricurvum* as the most frequently cited genera.[42] The SOB populations on the carrier were detected to be at a relatively low abundance among the bacteria community (2.7%). This was dominated by genus *Fastidiosipila* (24%), followed by *Proteiniphilum* (15%) and *Caldicoprobacter* (6%). In Table 5, genus *Sulfuricurvum*, belonging to *ε*-Proteobacteria, was identified with a higher relative abundance (0.96%) compared to other SOB genera, and it was dominant among the SOB members, followed by *Thiobacillus* (0.78%), *Acinetobactor*, belonging to *γ*-*Proteobacteria* (0.76%), and *Pseudomonas* (0.23%). *Sulfuricurvum* sp., classified as an autotrophic genus, was reported to grow facultatively, and this genus was capable of oxidizing sulfide, elemental sulfur, and thiosulfate while also causing the accumulation of elemental sulfur extracellularly.[43]As for the genus *Thiobacillus*, some members belonging to which was a chemolithotrophic SOB group, it has been commonly studied and reported to be the most appropriate microbes for sulfide removal due to its rate of sulfide oxidation, modest nutrition requirements, and higher affinity for sulfide and oxygen.[44] This SOB group could oxidize a variety of sulfur compounds. The genus *Thiobacillus* sp. and *Pseudomonas* sp. identified in this study could be responsible for the oxidation of the intermediate products of some reduced organic sulfur compounds such as H_2S.[45] In addition, these genera of SOB were also identified in the liquid phase of both digesters; the SOB population identified in R2 liquid phase may also contribute to the efficient sulfide removal in the liquid phase. As for the scarce existence of several genera of SOB in the control, it could be explained that some species of SOB such as *Thiobacillus thioparus* could grow under anaerobic conditions,

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with nitrate or nitrite as electron acceptors rather than oxygen.[46]

The bacterial community in the liquid phase of R1 and R2 is shown in Figures 7 and 8. The diversity of the bacterial

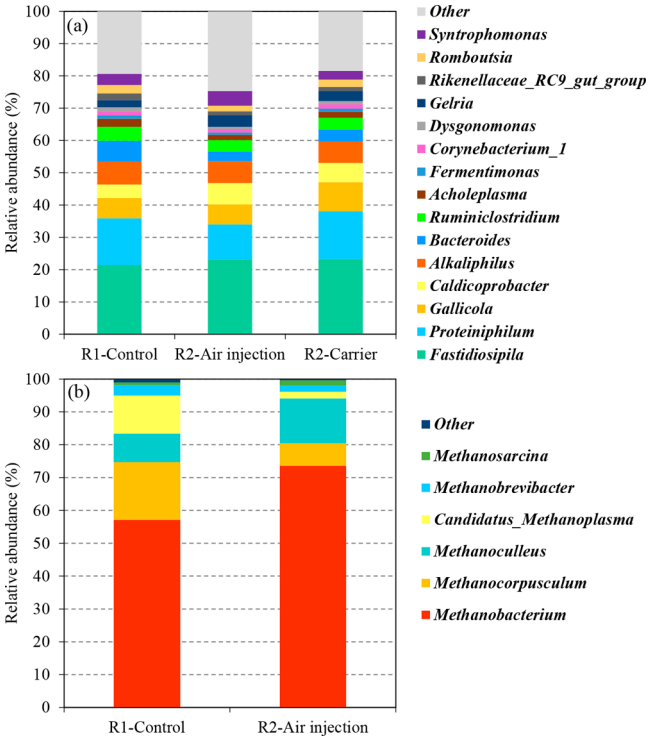


Figure 8. Microbial community structure at the genus level with top 15 relative abundance of bacteria (a) and archaea (b) on the 16S rDNA in the R1, R2 digester, and R2 carrier at the 120th day.

community in R2 was higher than R1, displaying a higher Shannon index (4.2 vs 3.9) and a lower Simpson index (0.04 vs 0.06) compared with the control group. The higher diversity in the microaerated digester has also been described by Lim et al. in 2014.[16] Figure 7 illustrates that phylum Firmicutes dominated R1 (57%) and R2 (64%) followed by the Bacteroidetes (22% and 25%) and Cloacimonetes (11% and 5%). Firmicutes is capable of producing several extracellular enzymes, that is, carbohydrase, protease, and lipases, which degrade different kinds of organic matter. The enrichment of Firmicutes population indicated the acceleration of hydrolysis and explained higher conversion rate obtained in section 3.2, supporting the hypothesis of the air supplied digester being a more favorable habitat for hydrolytic and acidogenic bacteria and thus accelerating the metabolism of more complex substrates. Accordingly, in Figure 8a, the relative abundance of facultative bacteria, genera *Fastidiosipila*, able to convert carbohydrate and protein to acetate, was observed to increase remarkably in R2 compared to that in R1. Also, *Gelria*, a hydrogen-producing bacteria, was found to be higher in R2 (3.9%) compared to R1 (2.0%). The relative abundance of genera *Syntrophmonas* increased in R2 (4.8%) compared to R1 (3.2%), which is capable of oxidizing acetate and thus able to work in a syntrophic association with the hydrogenotrophic methanogens.[47] The results of the bacterial community indicated that limited air injection may facilitate the enrichment of facultatively hydrolytic and syntrophic bacteria, which could make the hydrolysis in the overall fermentation process more efficient.

3.6. Methanogenic Community Structure Changes with Air Injection. The archaeal diversity in R2 was found to be obviously lower than that of the control digester (R1). More specifically, R2 exhibited a lower Shannon index (1.05) and a higher Simpson index (5.0) compared to that attained in R1 (1.35 and 3.0, respectively). Air injection was found to affect the archaeal community structure.[17]

In Figure 8b, the hydrogenotrophic methanogens *Methanobacterium* are shown to dominate the archaeal community of R2, accounting for 75% of all reads followed by *Methanoculleus* (14%) and *Methanocorpusculum* (6%). The relative abundances of these three archaea were 57%, 8%, and 17% in the R1 without air supplement. The raised percentage of *Methanoculleus* in R2 was in accordance with Tang et al. in 2004,[18] who reported a significant increment of the *Methanoculleus* group under microaeration conditions. In addition, *Methanosarcina* spp which were acetate and hydrogen mix-trophic methanogen,[48] approximately doubled to 1.6% in R2. The increased relative abundances of hydrogenotrophic methanogens may be attributed to following two reasons. First, the enrichment of syntrophic bacteria (*Syntrophomonas*) as discussed in section 3.5 accelerated the synergetic interaction between hydrogen-producing bacteria and hydrogenotrophic methanogens. Second, the lower abundance of H_2-utilized SRB, their potential competitors, such as *Desulfurovibrio* and *Desulfobulbus*, provided a more suitable habitat for methanogens to function. The enhanced hydrogenotrophic methanogenesis with air injection could help maintain lower hydrogen partial pressure and promote the AD stability.

4. CONCLUSION

Limited air supplement achieved high H_2S removal efficiency (>99%) while also enhancing the methane production from chicken manure AD. Air injections led to lower levels of VFAs and higher soluble iron content, which explained the improved AD process performance. The functions of the H_2-utilizing SRB group were restrained, whereas functional SOB members were strengthened in the air supplied digester. Meanwhile, higher bacterial diversity, an increment in Firmicutes members, and hydrogentrophic methanogens, *Methanobacterium* and *Methanoculleus*, explained the improved stability of the process performance. This study indicated that in situ desulfurization technology is a feasible candidate for efficient H_2S removal in a conventional biogas plant.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Owamah, H. I.; Alfa, M. I.; Dahunsi, S. O. Optimization of biogas from chicken droppings with Cymbopogon citratus. *Renewable Energy* **2014**, *68*, 366−371.

(2) Bi, S.; Qiao, W.; Xiong, L.; Mahdy, A.; Wandera, S. M.; Yin, D.; Dong, R. Improved high solid anaerobic digestion of chicken manure by moderate in situ ammonia stripping and its relation to metabolic pathway. *Renewable Energy* **2020**, *146*, 2380−2389.

(3) Mahdy, A.; Bi, S.; Song, Y.; Qiao, W.; Dong, R. Overcome inhibition of anaerobic digestion of chicken manure under ammonia-stressed condition by lowering the organic loading rate. *Bioresource Technology Reports.* **2020**, *9*, 100359.

(4) Kobayashi, T.; Li, Y.; Kubota, K.; Harada, H.; Maeda, T.; Yu, H. Characterization of sulfide-oxidizing microbial mats developed inside a full-scale anaerobic digester employing biological desulfurization. *Appl. Microbiol. Biotechnol.* **2012**, *93* (2), 847−857.

(5) Ramos, I.; Fdz-Polanco, M. The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: results from a pilot-scale digester treating sewage sludge. *Bioresour. Technol.* **2013**, *140*, 80−85.

(6) Ciahotný, K.; Kyselová, V. Hydrogen Sulfide Removal from Biogas Using Carbon Impregnated with Oxidants. *Energy Fuels* **2019**, *33* (6), 5316−5321.

(7) Khoshnevisan, B.; Tsapekos, P.; Alfaro, N.; Diaz, I.; Fdz-Polanco, M.; Rafiee, S.; Angelidaki, I. A review on prospects and challenges of biological H2S removal from biogas with focus on bio trickling filtration and microaerobic desulfurization. *Biofuel Research Journal.* **2017**, *4* (4), 741−750.

(8) Duangmanee, T. Micro-aeration for hydrogen sulfide removal from biogas. Dissertations & Theses, Gradworks, 2009.

(9) Schneider, R. L.; Quicker, P.; Anzer, T.; Prechtl, S.; Faulstich, M. Grundlegende Untersuchungen zur effektiven, kostengünstigen Entfernung von Schwefelwasserstoff aus Biogas. In *Biogasanlagen Anforderungen zur Luftreinhaltung*, Ausburg, 2002.

(10) Ikbal; Tang, Y.; Shigematsu, T.; Morimura, S.; Kida, K. Methanogenic activity and repression of hydrogen sulfide evolved during high rate thermophilic methane fermentation of municipal solid waste. *Jpn. J. Water Treat Biol.* **2003**, *39* (1), 17−24.

(11) Mahdy, A.; Song, Y.; Salama, A.; Qiao, W.; Dong, R. Simultaneous H_2S mitigation and methanization enhancement of chicken manure through the introduction of the micro-aeration approach. *Chemosphere* **2020**, *253*, 126687.

(12) Chen, Q.; Wu, W.; Qi, D.; Ding, Y.; Zhao, Z. Review on micro-aeration-based anaerobic digestion: State of the art, challenges, and prospective. *Sci. Total Environ.* **2020**, *710*, 136388.

(13) Mulbry, W.; Selmer, K.; Lansing, S. Effect of liquid surface area on hydrogen sulfide oxidation during micro-aeration in dairy manure digesters. *PLoS One* **2017**, *12* (10), e0185738.

(14) Botheju, D.; Samarakoon, G.; Chen, C.; Bakke, R. An Experimental Study on the Effects of Oxygen in Bio-Gasification; Part 1. In *Proceedings of the International Conference on Renewable Energies and Power Quality (ICREPQ 10)*; Granada: Spain, 2010.

(15) Cai, Y.; Hu, K.; Zheng, Z.; Zhang, Y.; Guo, S.; Zhao, X.; Cui, Z.; Wang, X. Effects of adding EDTA and Fe^{2+} on the performance of reactor and microbial community structure in two simulated phases of anaerobic digestion. *Bioresour. Technol.* **2019**, *275*, 183−191.

(16) Lim, J. W.; Chiam, J. A.; Wang, J. Microbial community structure reveals how micro-aeration improves fermentation during anaerobic co-digestion of brown water and food waste. *Bioresour. Technol.* **2014**, *171* (1), 132−138.

(17) Nguyen, D.; Khanal, S. K. A little breath of fresh air into an anaerobic system: how micro-aeration facilitates anaerobic digestion process. *Biotechnol. Adv.* **2018**, *36*, 1971−1983.

(18) Tang, Y.; Shigematsu, T.; Ikbal; Morimura, S.; Kida, K. The effects of micro-aeration on the phylogenetic diversity of micro-organisms in a thermophilic anaerobic municipal solid-waste digester. *Water Res.* **2004**, *38*, 2537−2550.

(19) Fu, S.-F.; Wang, F.; Shi, X.-S.; Guo, R.-B. Impacts of micro-aeration on the anaerobic digestion of corn straw and the microbial community structure. *Chem. Eng. J.* **2016**, *287*, 523−528.

(20) Pokorna, D.; Carceller, J. M.; Paclik, L.; Zabranska, J. Biogas cleaning by hydrogen sulfide scrubbing and bio-oxidation of captured sulfides. *Energy Fuels* **2015**, *29* (7), 4058−4065.

(21) Nghiem, L. D.; Manassa, P.; Dawson, M.; Fitzgerald, S. K. Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas. *Bioresour. Technol.* **2014**, *173*, 443−447.

(22) Ruan, R.; Cao, J.; Li, C.; Zheng, D.; Luo, J. The influence of micro-oxygen addition on desulfurization performance and microbial communities during waste-activated sludge digestion in a rusty scrap iron-loaded anaerobic digester. *Energies* **2017**, *10* (2), 258.

(23) Wandera, S. M.; Qiao, W.; Algapani, D. E.; Bi, S.; Yin, D.; Qi, X.; Liu, Y.; Dach, J.; Dong, R. Searching for possibilities to improve the performance of full scale agricultural biogas plants. *Renewable Energy* **2018**, *116*, 720−727.

(24) APHA (American Public Health Association). *Standard Methods for the Examination of Water and Wastewater*, 20th ed.; Washington, DC, 2005.

(25) Bi, S.; Westerholm, M.; Qiao, W.; Mahdy, A.; Xiong, L.; Yin, D.; Fan, R.; Dach, J.; Dong, R. Enhanced methanogenic performance and metabolic pathway of high solid anaerobic digestion of chicken manure by Fe^{2+} and Ni^{2+} supplementation. *Waste Manage.* **2019**, *94*, 10−17.

(26) Abatzoglou, N.; Boivin, S. A review of biogas purification processes. *Biofuels, Bioprod. Biorefin.* **2009**, *3* (1), 42−71.

(27) Botheju, D.; Bakke, R. Oxygen effects in anaerobic digestion − a review. *Open Waste Manage. J.* **2011**, *4*, 1−19.

(28) Appels, L.; Baeyens, J.; Degrève, J.; Dewil, R. Principles and potential of the anaerobic digestion of waste-activated sludge. *Prog. Energy Combust. Sci.* **2008**, *34* (6), 755−781.

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(29) Johansen, J. E.; Bakke, R. Enhancing hydrolysis with micro-aeration. *Water Sci. Technol.* **2006**, *53* (8), 43−50.
(30) Rodriguez, E.; Lopes, A.; Fdz-Polanco, M.; Stams, A.; Garcia-Encina, P. Molecular analysis of the biomass of a fluidized bed reactor treating synthetic vinasse at anaerobic and micro-aerobic conditions. *Appl. Microbiol. Biotechnol.* **2012**, *93* (5), 2181−2191.
(31) Lim, J. W.; Wang, J. Enhanced hydrolysis and methane yield by applying micro- aeration pretreatment to the anaerobic co-digestion of brown water and food waste. *Waste Manage.* **2013**, *33*, 813−819.
(32) Tsapekos, P.; Kougias, P. G.; Vasileiou, S. A.; Lyberatos, G.; Angelidaki, I. Effect of micro-aeration and inoculum type on the biodegradation of lignocellulosic substrate. *Bioresour. Technol.* **2017**, *225*, 246−253.
(33) Wang, G.; Li, Q.; Gao, X.; Wang, X. Sawdust-derived biochar much mitigates VFAs accumulation and improves microbial activities to enhance methane production in thermophilic anaerobic digestion. *ACS Sustainable Chem. Eng.* **2019**, *7* (2), 2141−2150.
(34) Fu, S.; Wang, F.; Yuan, X.; Yang, Z.; Luo, S.; Wang, C.; Guo, R. The thermophilic (55°C) microaerobic pretreatment of corn straw for anaerobic digestion. *Bioresour. Technol.* **2015**, *175*, 203−208.
(35) Zhou, W.; Imai, T.; Ukita, M.; Li, F.; Yuasa, A. Effect of limited aeration on the anaerobic treatment of evaporator condensate from a sulfite pulp mill. *Chemosphere* **2007**, *66* (5), 924−929.
(36) Giordano, A.; Di Capua, F.; Esposito, G.; Pirozzi, F. Long-term biogas desulfurization under different microaerobic conditions in full-scale thermophilic digesters co-digesting high-solid sewage sludge. *Int. Biodeterior. Biodegrad.* **2019**, *142*, 131−136.
(37) Wang, J.; Westerholm, M.; Qiao, W.; Mahdy, A.; Wandera, S. M.; Yin, D.; Bi, S.; Fan, R.; Dong, R. Enhancing anaerobic digestion of dairy and swine wastewater by adding trace elements: evaluation in batch and continuous experiments. *Water Sci. Technol.* **2019**, *80* (9), 1662−1672.
(38) Yun, S.; Zhang, C.; Wang, Y.; Zhu, J.; Huang, X.; Du, T.; Li, X.; Wei, Y. Synergistic effects of Fe salts and composite additives on anaerobic digestion of dairy manure. *Int. Biodeterior. Biodegrad.* **2019**, *136*, 82−90.
(39) Hao, T.-w.; Xiang, P.-y.; Mackey, H. R.; Chi, K.; Lu, H.; Chui, H.-k.; van Loosdrecht, M. C.M.; Chen, G.-H. A review of biological sulfate conversions in wastewater treatment. *Water Res.* **2014**, *65*, 1−21.
(40) Stackebrandt, E.; Sproer, C.; Rainey, F. A.; Burghardt, J.; Pauker, O.; Hippe, H. Phylogenetic analysis of the genus desulfotomaculum: evidence for the misclassification of desulfotomaculum guttoideum and description of desulfotomaculum orientis as desulfosporosinus orientis gen. nov. comb. nov. *Int. J. Syst. Bacteriol.* **1997**, *47* (4), 1134−1139.
(41) Jenicek, P.; Celis, C. A.; Koubova, J.; Pokorna, D. Comparison of microbial activity in anaerobic and microaerobic digesters. *Water Sci. Technol.* **2011**, *63* (10), 2244−2249.
(42) Krayzelova, L.; Bartacek, J.; Diaz, I.; Jeison, D.; Volcke, E.; Jenicek, P. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. *Rev. Environ. Sci. Bio/Technol.* **2015**, *14* (4), 703−725.
(43) Handley, K. M.; Bartels, D.; O'Loughlin, E. J.; Williams, K. H.; Trimble, W. L.; Skinner, K.; Gilbert, J. A.; Desai, N.; Glass, E. M.; Paczian, T.; Wilke, A.; Antonopoulos, D.; Kemner, K. M.; Meyer, F. The complete genome sequence for putative H2- and S-oxidizer candidatus *Sulfuricurvum* sp. assembled de novo from an aquifer-derived metagenome. *Environ. Microbiol.* **2014**, *16* (11), 3443−3462.
(44) Janssen, A. J.H.; Lens, P. N.L.; Stams, A. J.M.; Plugge, C. M.; Sorokin, D. Y.; Muyzer, G.; Dijkman, H.; Van Zessen, E.; Luimes, P.; Buisman, C. J.N. Application of bacteria involved in the biological sulfur cycle for paper mill effluent purification. *Sci. Total Environ.* **2009**, *407* (4), 1333−1343.
(45) Ito, T.; Miyaji, T.; Nakagawa, T.; Tomizuka, N. Degradation of dimethyl disulfide by pseudomonas fluorescens strain 76. *Biosci., Biotechnol., Biochem.* **2007**, *71*, 366−370.
(46) Tang, K.; Baskaran, V.; Nemati, M. Bacteria of the sulphur cycle: an overview of microbiology, biokinetics and their role in petroleum and mining industries. *Biochem. Eng. J.* **2009**, *44* (1), 73−94.
(47) Chen, H.; Wang, W.; Xue, L.; Chen, C.; Liu, G.; Zhang, R. Effects of ammonia on anaerobic digestion of food waste: process performance and microbial community. *Energy Fuels* **2016**, *30* (7), 5749−5757.
(48) Cai, W.; Liu, W.; Yang, C.; Wang, L.; Liang, B.; Thangavel, S.; Guo, Z.; Wang, A. Biocathodic methanogenic community in an integrated anaerobic digestion and microbial electrolysis system for enhancement of methane production from waste sludge. *ACS Sustainable Chem. Eng.* **2016**, *4* (9), 4913−4921.

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